



ITSIMPLERA INSTITUTE

Network Administration Internship Program

WEEK 01 ASSIGNMENT REPORT

Networking Fundamentals & Network Simulation Tools

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1.Introduction

Computer networking forms the backbone of modern communication systems, enabling devices to exchange data efficiently across local and global infrastructures. Understanding these fundamentals is essential for every aspiring network administrator. This internship task, conducted under the supervision of the **Senior Network Administrator at IT-Simplera Institute**, focuses on building a strong foundation in networking concepts and gaining hands-on experience with industry-standard simulation tools — **Cisco Packet Tracer** and **GNS3**.

Through this activity, interns explore how networks operate, how devices communicate using IP addressing, and how connectivity is verified through commands like *ping*. By designing and configuring a simple Local Area Network (LAN) topology in both platforms, the task bridges theoretical learning with practical implementation. The objective is not only to understand the technical workflow but also to develop professional documentation and presentation skills that reflect real-world industry standards.

2.Networking Concepts Learned

During this internship task, several core networking concepts were studied and applied in practice. These concepts form the foundation of computer networking and were reinforced through hands-on simulation in Cisco Packet Tracer and GNS3.

1. **IP Addressing** Every device in a network requires a unique IP address to communicate. Understanding the distinction between network ID, broadcast address, and usable host addresses ensures proper configuration.
2. **Subnet Masks** Subnet masks define the boundary between the network and host portions of an IP address. For example, a subnet mask of `255.255.255.0` allows up to 254 usable host addresses in a single LAN.
3. **Switch Functionality** Switches operate at Layer 2 of the OSI model, forwarding frames based on MAC addresses. They provide connectivity between multiple devices in a Local Area Network (LAN).

4. **Static IP Configuration** Assigning static IP addresses ensures predictable communication between devices. In this lab, PCs were configured with IPs such as 192.168.1.2 and 192.168.1.3 to establish connectivity.
 5. **Ping Command** The ping utility was used to test end-to-end connectivity. Successful replies confirmed that devices were correctly configured and able to communicate within the LAN.
 6. **LAN Topology** A Local Area Network (LAN) is a network that connects devices within a limited area such as a lab, office, or campus. In this task, a simple LAN topology was designed using two PCs connected to a Cisco 2960 switch. This demonstrated how devices communicate through a central switch, how IP addresses enable logical identification, and how connectivity can be verified using ping tests.
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7. **Network Simulation Tools**
 - a. *Cisco Packet Tracer*: Provides a user-friendly, drag-and-drop interface for designing and testing networks.
 - b. *GNS3*: Offers advanced, console-based configuration using real device images, closely simulating real-world environments.

Summary Table

Concept	Key Learning	Example
IP Addressing	Logical identity of devices	PC1: 192.168.1.2
Subnet Mask	Defines network/host bits	255.255.255.0
Switch Functionality	Connects PCs via MAC addresses	Cisco 2960
Static IP	Manual assignment for predictable addressing	PC2: 192.168.1.3
Ping Command	Tests connectivity using ICMP	ping 192.168.1.3
LAN Topology	Simple design with PCs and switch	PC1 ↔ Switch ↔ PC2
Simulation Tools	Practice environments	Packet Tracer, GNS3

3. Cisco Packet Tracer Interface Overview

Cisco Packet Tracer is a powerful simulation tool that allows users to design, configure, and test computer networks in a virtual environment. Its interface is designed to be intuitive, enabling beginners to quickly build topologies while also offering advanced features for professional practice.

Key Components of the Interface

1. Device Palette (Left Panel)

- a. Contains categories of networking devices such as routers, switches, PCs, servers, and wireless equipment.
- b. Devices can be dragged and dropped into the workspace to build topologies.

2. Logical Workspace (Center Area)

- a. The main canvas where network devices are placed and interconnected.
- b. Supports both logical and physical views, allowing users to visualize how networks are structured.

3. Connection Toolbar (Bottom Panel)

- a. Provides different cable types (straight-through, crossover, fiber, serial) for connecting devices.
- b. Includes the **Place Note** tool for labeling devices, IP addresses, and ports, which enhances documentation clarity.

4. Simulation Controls (Bottom Right)

- a. Allows switching between **Realtime Mode** and **Simulation Mode**.
- b. Simulation mode enables packet flow visualization, helping users understand how data travels across the network.

5. Top Menu Bar

- a. Offers options for saving projects, configuring preferences, and accessing advanced features.

4. GNS3 Interface Overview

GNS3 (Graphical Network Simulator-3) is an advanced network simulation platform that allows users to design and test complex topologies using real device images. Unlike Packet Tracer, which is primarily educational, GNS3 provides a closer-to-real-world environment, making it widely used by professionals for certification preparation and lab testing.

Key Components of the Interface

1. Device Toolbar (Left Panel)

- a. Contains routers, switches, PCs, and other virtual devices.
- b. Devices can be dragged into the workspace to build topologies.

2. Topology Workspace (Center Area)

- a. The main canvas where devices are placed and interconnected.
- b. Supports large-scale designs with multiple routers, switches, and servers.

3. Console Window (Bottom Panel)

- a. Provides direct command-line access to devices.
- b. Used to configure IP addresses, routing protocols, and test connectivity.

4. Project Tabs (Top Bar)

- a. Allows multiple projects to be opened and managed simultaneously.
- b. Useful for organizing different lab scenarios.

5. Connection Tool

- a. Enables linking devices with Ethernet, serial, or other interfaces.
- b. Essential for building functional topologies.

5.Implementation of Topologies

A.LAN Topology in Packet Tracer

- Two PCs (PC0 and PC1) connected via Cisco 2960 switch using FastEthernet ports.
- PC0 → IP 192.168.1.2, Subnet 255.255.255.0.
- PC1 → IP 192.168.1.3, Subnet 255.255.255.0.
- Addressing started from .2 because .1 is usually reserved for the **default gateway (router)**.
- **Network Address:** 192.168.1.0 → represents the entire subnet.
- **Broadcast Address:** 192.168.1.255 → used to send packets to all devices in the LAN.
- **Usable Host Range:** 192.168.1.1 – 192.168.1.254.
- Connectivity verified with **ping command** (PC0 → PC1).

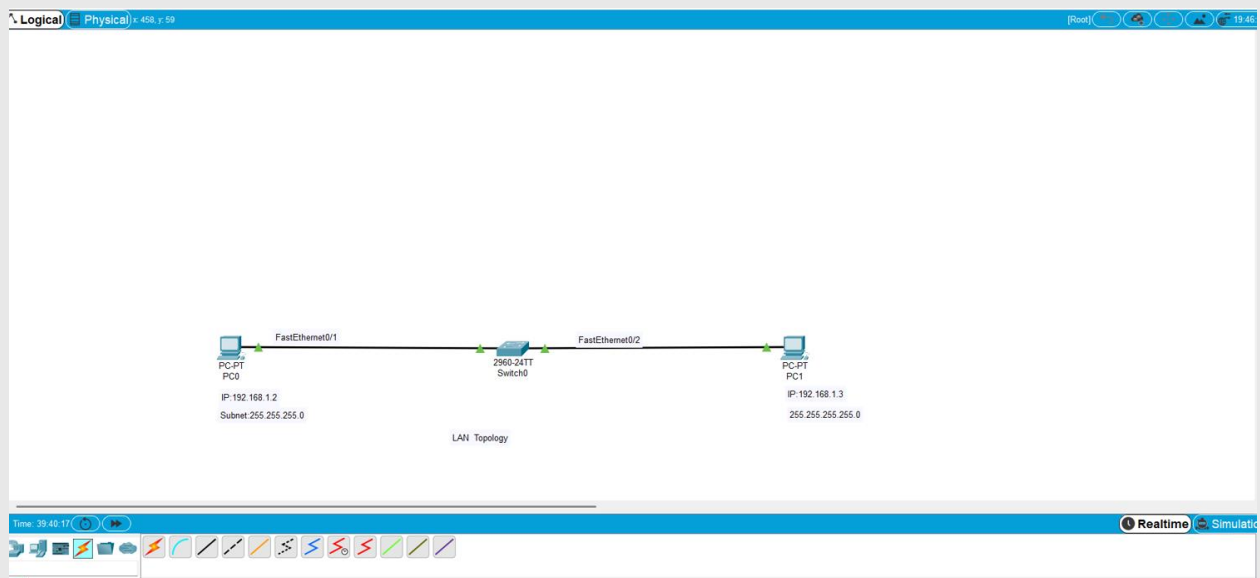
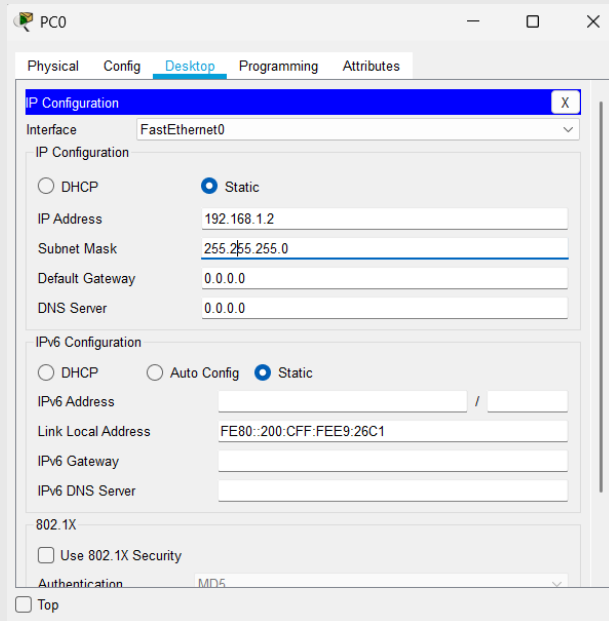


Figure: LAN topology in Cisco Packet Tracer showing two PCs connected to a Cisco 2960 switch with static IP configuration.

PC0, IP configuration



The screenshot shows the 'IP Configuration' window for PC0. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The 'IP Address' is 192.168.1.2, 'Subnet Mask' is 255.255.255.0, 'Default Gateway' is 0.0.0.0, and 'DNS Server' is 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is selected. The 'IPv6 Address' field is empty, 'Link Local Address' is FE80::200:CFF:FEE9:26C1, 'IPv6 Gateway' is empty, and 'IPv6 DNS Server' is empty. The '802.1X' section has 'Use 802.1X Security' unchecked and 'Authentication' set to MD5. A 'Top' button is at the bottom left.

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IP Address 192.168.1.2

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::200:CFF:FEE9:26C1

IPv6 Gateway

IPv6 DNS Server

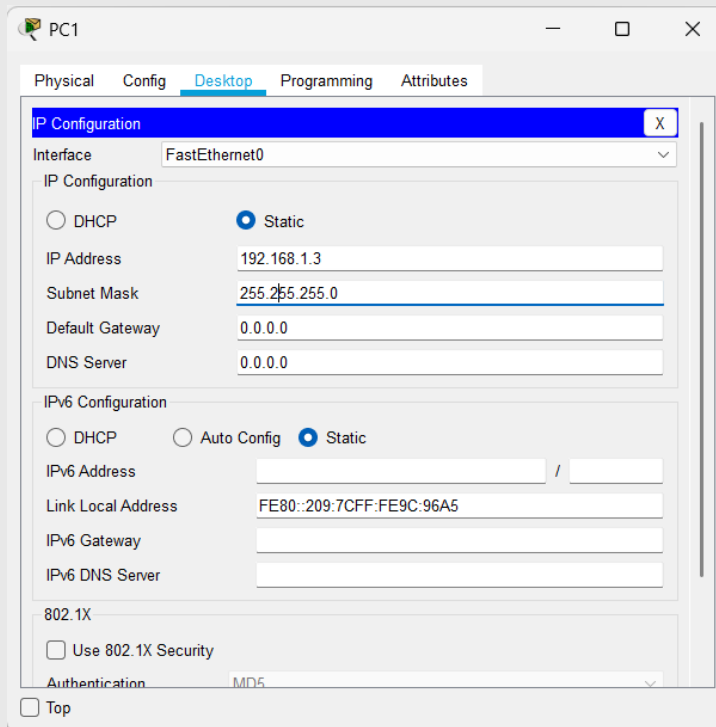
802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC1, IP Configuration



The screenshot shows the 'IP Configuration' window for PC1. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The 'IP Address' is 192.168.1.3, 'Subnet Mask' is 255.255.255.0, 'Default Gateway' is 0.0.0.0, and 'DNS Server' is 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is selected. The 'IPv6 Address' field is empty, 'Link Local Address' is FE80::209:7CFF:FE9C:96A5, 'IPv6 Gateway' is empty, and 'IPv6 DNS Server' is empty. The '802.1X' section has 'Use 802.1X Security' unchecked and 'Authentication' set to MD5. A 'Top' button is at the bottom left.

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IP Address 192.168.1.3

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::209:7CFF:FE9C:96A5

IPv6 Gateway

IPv6 DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

B.LAN TOPOLOGY in GNS3

- Two PCs (PC1 and PC2) connected via **Ethernet Switch**.
- IP addresses assigned:
 - PC1 → 192.168.1.2 / 255.255.255.0
 - PC2 → 192.168.1.3 / 255.255.255.0
- Addressing started from .2 because .1 is usually reserved for the **default gateway (router)**.
- **Network Address:** 192.168.1.0 → represents the entire subnet.
- **Broadcast Address:** 192.168.1.255 → used to send packets to all devices in the LAN.
- **Usable Host Range:** 192.168.1.1 – 192.168.1.254.
- Connectivity verified with **ping command** (PC1 → PC2).
- Green interface dots confirm active connections in GNS3 topology.

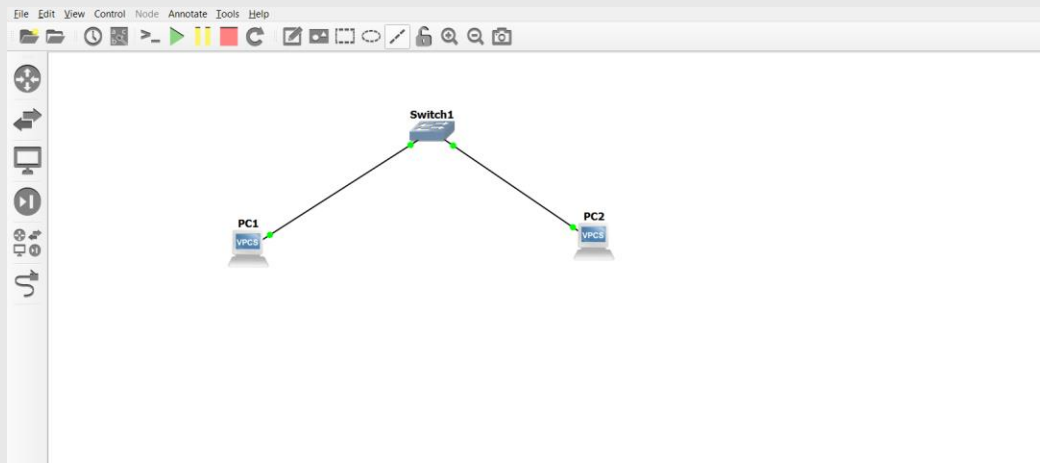
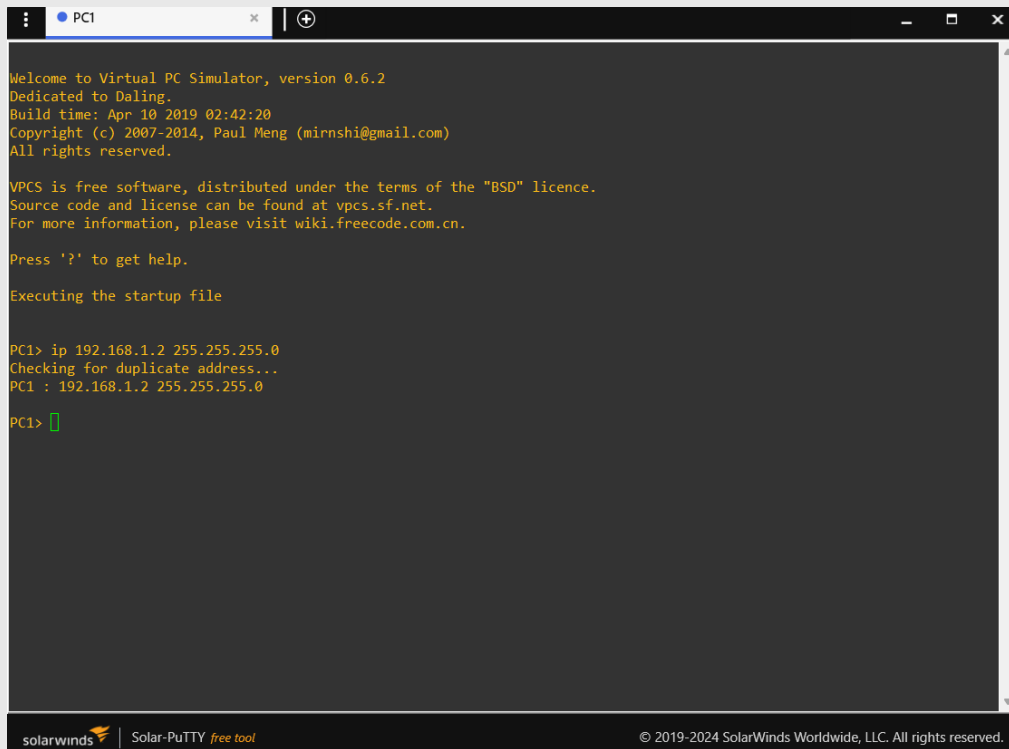


Figure: LAN topology in GNS3 showing two PCs connected to an Ethernet switch with static IP configuration.

PC1,IP Configuration



```

Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

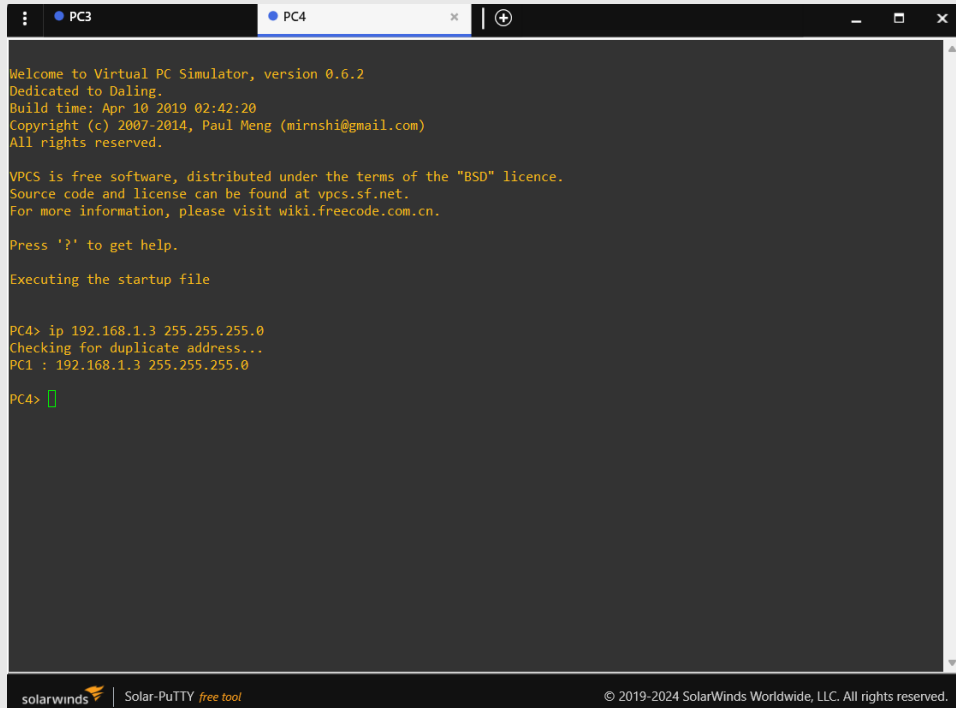
Executing the startup file

PC1> ip 192.168.1.2 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.2 255.255.255.0

PC1> 
```

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PC2,IP Configuration



The screenshot shows a SolarWinds Solar-PuTTY terminal window with two tabs: PC3 and PC4. The PC4 tab is active, displaying the following text:

```
Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

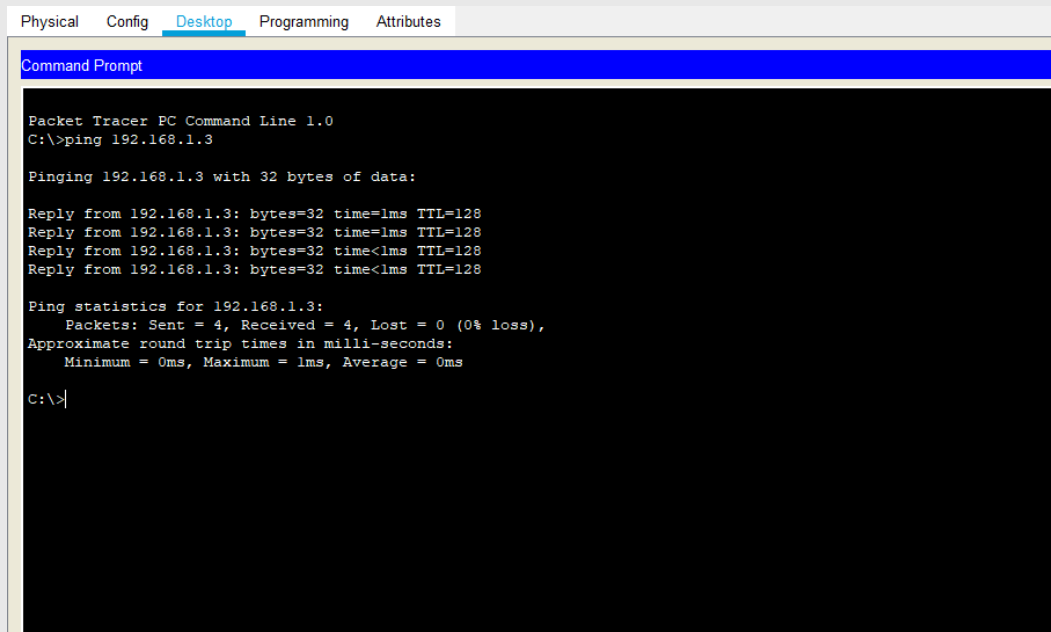
PC4> ip 192.168.1.3 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.3 255.255.255.0

PC4> 
```

The bottom of the window features a status bar with the SolarWinds logo, the text "Solar-PuTTY free tool", and the copyright notice "© 2019-2024 SolarWinds Worldwide, LLC. All rights reserved."

6. Ping Test Results

A. Ping test results in packet tracer



The screenshot shows the Packet Tracer interface with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the results of a ping test from 192.168.1.3 to 192.168.1.3. The output shows four successful replies with 0% loss and a 0ms average round trip time.

```
Physical  Config  Desktop  Programming  Attributes

Command Prompt

Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

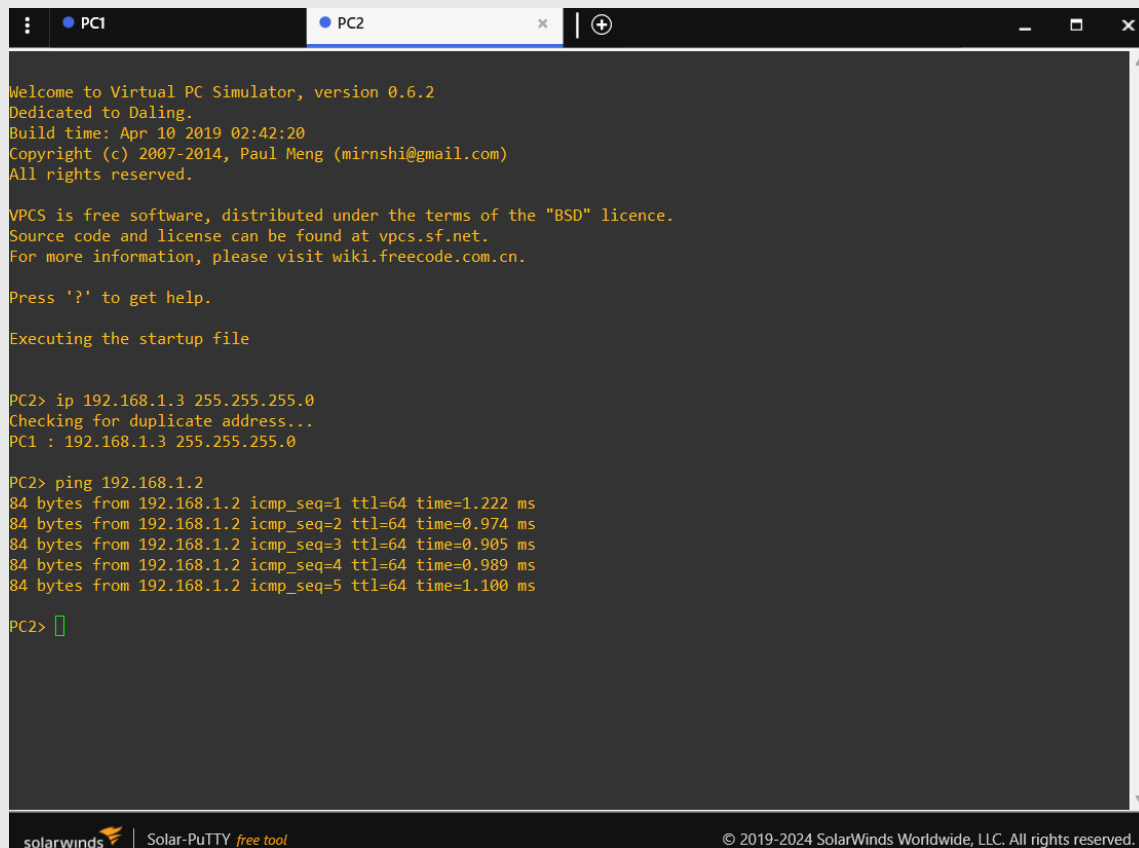
Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

B. Ping test results in GNS3



The screenshot shows a SolarWinds Solar-PuTTY terminal window with two tabs: PC1 and PC2. The PC2 tab is active, displaying the output of the VPCS startup and a ping test. The terminal text is as follows:

```
Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC2> ip 192.168.1.3 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.3 255.255.255.0

PC2> ping 192.168.1.2
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=1.222 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=0.974 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=0.905 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=0.989 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=1.100 ms

PC2> 
```

The terminal window has a dark background with yellow text. The SolarWinds logo and "Solar-PuTTY free tool" are visible in the bottom left, and the copyright notice "© 2019-2024 SolarWinds Worldwide, LLC. All rights reserved." is in the bottom right.

7.Challenges Faced in GNS3 LAN Simulation

- **Dependency Errors:** Encountered missing `wpcap.dll` error while dragging Ethernet switch, requiring Npcap installation in WinPcap-compatible mode.
- **Software Compatibility:** GNS3 required administrator privileges and proper driver installation to run smoothly.
- **Configuration Complexity:** Unlike Packet Tracer, IP addressing in GNS3 VPCS had to be set manually using console commands.
- **Troubleshooting Time:** Multiple reinstallations (Npcap/GNS3) were needed to resolve persistent errors.
- **Learning Curve:** Understanding differences between Packet Tracer (GUI-based) and GNS3 (realistic emulation) took extra effort.
- **Resource Usage:** GNS3 consumed more system resources compared to Packet Tracer, sometimes slowing down the simulation.

- **Connectivity Verification:** Ensuring successful ping between PCs required careful IP assignment and subnet mask configuration.

8. Conclusion

- The LAN topology was successfully designed and implemented in both **Cisco Packet Tracer** and **GNS3**.
- Static IP addressing ensured proper communication between PCs, with successful **ping tests** verifying connectivity.
- GNS3 provided a more realistic environment by requiring manual configuration and external dependencies (Npcap), while Packet Tracer offered a simpler GUI-based setup.
- Challenges such as missing `wpcap.dll` and driver compatibility were resolved through troubleshooting, enhancing practical learning.
- The exercise strengthened understanding of **network fundamentals**, including IP addressing, subnetting, and switch-based communication.
- Overall, the simulation demonstrated how LANs operate in real-world scenarios and highlighted the differences between educational tools and professional emulation platforms.